

Eventless Payload-Adaptive NMPC for Quadrotors

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Abstract—A release command does not guarantee physical payload detachment, so switching a controller’s model from an attach/drop signal can be unsafe. We present an eventless payload-adaptive control framework in which a moving horizon estimator (MHE) [1] uses only motor speeds and odometry to publish an atomic, health-gated frame of composite mass, first mass moment $s = m_P r_{xy}$, and inertia to an outer-loop nonlinear model predictive controller (NMPC). The first-moment representation remains well defined as the payload vanishes, while bounded filtering and continuous body-rate scheduling prevent discontinuous model updates. A thrust-residual self-trigger deweights stale pre-change stages in the MHE window and reduces mass-estimate settling time by up to $4\times$. In PX4 [2] software-in-the-loop experiments with physically attached eccentric payloads, the eventless interface completes release confirmation in 21/21 valid paired flights, with no observed post-release instability, at a median confirmation delay of 3.76 s when the moment vote clears in maneuver and 16.46 s when a brake to hover is first required. A safety ablation shows that payload quantity and thrust residual are individually unreliable during aggressive flight: their fast release branch causes premature model clearing in 3/9 flights. Restricting the estimator’s release detector to fresh moment-collapse evidence eliminates premature declarations in a 12-flight smoke test, yielding 11 confirmations and one conservative unresolved outcome. Against an L1-adaptive [3] NMPC baseline, the proposed method provides comparable transient tracking, recovers 15% to 35% faster, and retains interpretable physical parameters. These results demonstrate the eventless control interface and its fail-safe semantics, while not claiming distribution-wide robustness of the release detector.

I. INTRODUCTION

Autonomous aerial delivery and in-flight aerial manipulation subject a multirotor to abrupt changes in its own dynamics. Grasping or releasing a parcel changes three coupled quantities at once: the total *mass* m (which rescales the thrust-to-acceleration map and the hover throttle), the *center of mass* (a thrust that no longer passes through the composite CoM produces a constant trim torque $\tau \approx \Delta r \times f$, significant even for small offsets because $f \approx mg$ is large), and the *rotational inertia* (which slows the attitude response and effectively lowers the inner-loop bandwidth). The first channel is an outer-loop (position/thrust) problem; the latter two are inner-loop (attitude/rate) problems. A controller that models the payload only as a scalar mass will mistrack or, for sufficiently heavy eccentric loads, destabilize.

Three families of methods address load changes. *Explicit parameter estimation* (EKF/MHE of mass, CoM, inertia) is interpretable, integrates natively with a model-based controller, and can update thrust-margin constraints, but classical estimators converge slowly across an abrupt change because the estimation window or filter averages across the discontinuity. *Lumped-disturbance adaptation* (L1-adaptive NMPC, INDI)

compensates the aggregate effect without estimating physical parameters, reacting fast but yielding no interpretable model. *Learning-based* methods (e.g. neural MHE that regresses estimator weights) can resolve the estimator’s bias–variance trade-off but typically require rich sensing or offline ground-truth data.

The central question of this work is whether the complete grasp–carry–release lifecycle can adapt its payload model without trusting a commanded or externally supplied payload event. Our answer rests on two observations. First, an abrupt payload change leaves a detectable common-mode signature in reconstructed collective thrust, so the estimator can discover *when* the dynamics changed without consuming a gripper command, attachment state, or external event notification. Second, **motor speeds are a free physical sensor**: both collective thrust and body torques can be reconstructed from rotor speeds alone, yielding physical residuals that are decoupled from the estimator’s own state. Combining the two yields an estimator that is both fast (it deweights stale pre-event measurements the moment the thrust residual crosses a threshold) and *sensing-minimal* (it needs no external event signal and no measurement of the attachment geometry).

Contributions.

- 1) An **eventless payload-adaptive NMPC architecture** whose model never consumes an attach/drop notification or treats its own release command as evidence. An atomic, health-gated $m/s/J$ frame drives bounded model updates and continuous body-rate scheduling (§III–IV).
- 2) A **self-triggered MHE** that detects abrupt payload changes from reconstructed thrust and deweights stale pre-event window stages, reducing mass-estimate settling time by up to $\sim 4\times$ without an external event signal. The first-moment state $s = m_P r_{xy}$ represents eccentricity without measured attachment geometry and remains well-defined at release (§IV; §VI-A).
- 3) **Closed-loop lifecycle and safety validation**: 21/21 valid paired flights complete eventless release without observed post-release instability. A targeted ablation rejects an unsafe quantity-plus-residual branch that lacks moment evidence and adds an UNRESOLVED fail-safe state (§VI-C).

The runtime estimator consumes only motor speeds and odometry; the sole geometric quantity not identified online is a weak vertical-arm prior r_z , which is structurally unobservable from the thrust-moment channel.

II. RELATED WORK

Agile quadrotor flight. Modern quadrotor control builds on geometric tracking on SE(3) [4], and pushing to the actuator limits is by now standard in time-optimal flight [5]. Both assume the inertial parameters are known and fixed; a payload event invalidates that assumption mid-flight, which is the gap this work addresses.

Aerial manipulation and transport. Aerial robots that physically interact with their environment—grasping, carrying, and releasing objects—are surveyed in [6]; dynamic aerial grasping, where the gripper dynamics must be explicitly modeled and controlled, was demonstrated in [7]. Load transportation specifically—by grasping or by cable suspension—is surveyed in [8]. Our setting is the transport-and-release subproblem, where the payload is rigidly attached and the resulting change in mass, inertia, and center of mass must be handled by the controller.

Control under changing payload. Adaptive and robust designs—L1-adaptive NMPC [9], INDI [10]—treat the load as a lumped disturbance and are the standard baselines for payload-carrying flight; L1-NMPC in particular reports large tracking-error reductions under unknown load on aggressive trajectories. A parallel line learns the residual dynamics instead: [11] fits unmodeled aerodynamic effects with Gaussian processes inside the MPC, which improves tracking without yielding a named physical parameter either. Slung-load control addresses the swinging-payload case explicitly [12], exploiting differential flatness of the cable-suspended system; a rigidly attached payload instead changes the vehicle’s own inertial parameters. A related proprioceptive line estimates the external wrench directly with momentum-based observers, enabling collision detection and interaction control [13]. All of these recover a *force/torque*, not a physical parameter: they cannot, e.g., update a thrust-margin constraint or an inner-loop gain schedule from the estimate.

MHE and online parameter estimation. Filter- and optimization-based estimators recover mass, inertia, and geometric parameters [14]–[16]. Standard MHE performance is highly sensitive to the process/measurement weight matrices; neural MHE (NeuroMHE) [17] learns to output these weights online, resolving the bias–variance trade-off but estimating a lumped disturbance force rather than explicit named parameters.

Event-triggered and change detection. Event- and self-triggered control schedule computation by state-dependent conditions rather than by a fixed clock [18]; our trigger is of the self-triggered kind in that the condition is evaluated from signals the estimator already has. Residual-based fault/change detection (CUSUM and related [19]) underlies it. Our positioning: we do not learn dynamics and add no sensing; every deployed signal derives from motor speeds and odometry. This is a deliberate departure from learning-based MPC [20] and the broader safe-learning agenda [21], where a learned component absorbs the model mismatch; here the mismatch is resolved into named parameters that downstream constraints

can consume. We separately test, as a supporting ablation, whether numerical tuning improves the handcrafted transient-period schedule.

III. SYSTEM OVERVIEW AND PROBLEM SETUP

Architecture (Fig. 1). Three layers: Gazebo physics (with a magnetic DetachableJoint gripper producing genuine inertia/CoM changes), the PX4 inner-loop rate controller, and a ROS 2 outer loop (NMPC + MHE). The outer loop publishes a collective thrust and a body-rate setpoint; PX4 tracks the body rate.

Dynamics and parameterization. The NMPC model carries the runtime parameter vector $p = [x_{\text{ref}}, m, \Delta J, c_{xy}]$ (extended with a lumped disturbance only for the L1 baseline). Thrust acting at the geometric center about the composite CoM produces $\tau_{\text{com}} = [-c_y T, +c_x T, 0]$. The deployed MHE augments both composite mass and the horizontal first mass moment,

$$x_{\text{MHE}} = [x; m; s_x; s_y] \in \mathbb{R}^{16}, \quad s = mPr_{xy}, \quad (1)$$

with $\dot{m} = 0$ and $\dot{s} = 0$ within each window ($N = 20$, $\Delta t = 0.1$ s). The published geometry is reconstructed as

$$c_{xy} = \frac{s}{m}, \quad \mu = \frac{m_B \max(m - m_B, 0)}{m}, \quad (2)$$

$$\Delta J_{xx} = \Delta J_{yy} = \mu r_z^2.$$

where r_z is a weak rack prior. The small $O(r_{xy}^2)$ diagonal terms and yaw increment are omitted in the deployed interface. The hover-input baseline is recomputed from the live model and includes the eccentric trim torque, avoiding a stale operating-point bias.

Model-independent physical inputs. We reconstruct the collective thrust $T_{\text{phys}} = k \sum_i \omega_i^2$ and the body torques from rotor speeds and arm geometry. The quadratic coefficient k is a static-thrust constant; it is known to lose validity under large induced inflow during fast translation [22], which we do not model and which is one source of the estimator bias discussed in Sec. VII. Feeding the NMPC’s *commanded* input back into the estimator instead creates a self-consistent blind spot: at hover $m \approx T/g$ is self-consistent for any m , so the mass is unobservable from the commanded thrust. The motor-speed reconstruction breaks this because it is decoupled from the controller/estimator state.

Thrust-to-throttle inversion. The offboard collective-thrust command is inverted to a normalized rotor command through the motor model $T = k\omega^2$; because k appears on both the forward and inverse paths, steady-state tracking is independent of the absolute thrust calibration.

Continuous inner-loop scheduling. Model consistency in the NMPC cannot by itself restore the PX4 rate loop’s lost bandwidth. The deployed path therefore maps the live inertia increment to a body-rate-command scale, low-pass filters it, applies target hysteresis and asymmetric rise/recovery slew limits, and preserves 5% command headroom before the final saturation. It does *not* rewrite PX4 parameters. An attach-time inertia-only bootstrap uses the rack’s 0.30 kg payload envelope

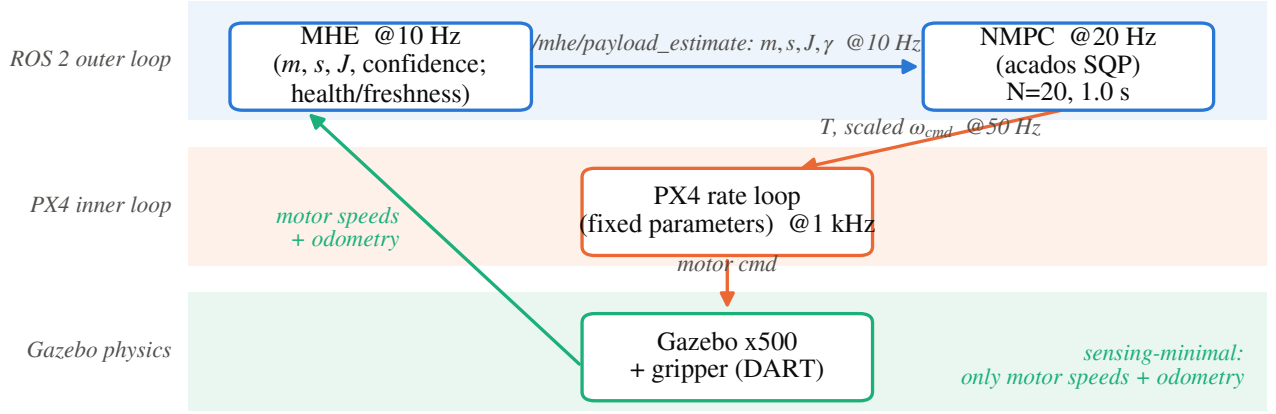


Fig. 1. System architecture and information flow. The full pipeline consumes only motor speeds and odometry; one atomic, health-gated $m/s/J$ frame drives both the NMPC model and continuous rate-command scheduling. No attach/drop notification enters this path.

for immediate safety, then fades out after persistent coherent MHE evidence; neither payload mass nor s is injected by this bootstrap, and the drop path remains estimate-only.

IV. SELF-TRIGGERED ESTIMATION AND EVENTLESS ADAPTATION

A. Deweighting mechanism

Within the sliding window, upon a payload event we deweight the R/Q blocks of the pre-event stages (per-stage `cost_set`), rather than softening the mass anchor Q_0 (which is already soft and is not the bottleneck; the true bottleneck is the $\dot{m} = 0$ rigidity that lets stale cross-discontinuity measurements pull the mass). Guardrails: the terminal measurement never enters the cost, and the first solve after an event keeps nominal weights.

B. No-signal self-trigger

The detector arms only after the baseline has settled (a warmup gate of consecutive quiet thrust-residual frames), then fires when $|T_{\text{phys}} - \text{baseline}|$ exceeds a threshold for a persistence count of consecutive frames. The warmup gate rejects the takeoff transient; the persistence count rejects gust pulses (broadband/transient) while catching a sustained mass step. This same criterion is the false-trigger criterion characterized under wind in §VI. This detector schedules the MHE window weights. Release confirmation uses the multi-source logic below; the two roles must not be conflated, because a threshold suitable for a 0.30 kg schedule event can suppress the residual vote of a 0.15 kg release.

C. Continuous payload interface and autonomous lifecycle

The MHE publishes one atomic frame

$$\mathcal{P} = \{m, s_x, s_y, c_x, c_y, \Delta J_{xx}, \Delta J_{yy}, \Delta J_{zz}, J_{xx}, J_{yy}, J_{zz}, \gamma, h, a\}, \quad (3)$$

where γ is no-payload confidence, h is a health bit and a is solution age. Health requires a successful latest solve and

fresh odometry, motor-speed, and known-input streams. The NMPC additionally checks local receive age; an unhealthy or stale frame freezes the last accepted model and breaks the confidence persistence timer. Valid frames pass through a low-pass filter and separate slew limits for m , s , and ΔJ before every NMPC solve.

Payload presence is an estimator-owned state, deliberately distinct from the legacy flag that means “an external attach notification was received.” A hysteretic `EMPTY` ↔ `LOADED` machine uses sustained payload-mass evidence, with a directional residual as a fast path, and a separate latch records that the present flight reliably entered `LOADED`. This latch prevents an empty-flight confidence value from consuming the later drop confirmation and prevents a transient mass dip during a figure-eight from disarming the true release path.

No-payload confidence is the product of smooth empty scores for payload quantity, first moment, and inertia. A nonzero moment therefore vetoes a false empty decision even if the mass estimate touches its lower bound. On accepted release evidence, the estimator switches only the *published target* of s to zero; the optimization state is not overwritten. The output filter consequently produces a continuous decay rather than a geometric step. The controller marks a planned drop complete, resets its L1 state, and finishes recovery only after healthy confidence remains above 0.90 for 1.0 s; issuing the gripper command itself does none of these things.

D. Dimensionless trigger schedule

The deployed hand-designed schedule M0 applies a fixed pre-event deweighting rhythm and expresses the confirmation threshold as the dimensionless ratio α , with physical threshold $\alpha g m_{\text{env}}$ normalized by the rack payload envelope rather than the unknown payload ($\alpha = 1.5$, $m_{\text{env}} = 0.30$ kg: 4.4 N in the eventless batches; the historical ablations used 1.5 N). It therefore does not scale with the actual payload. A supporting simulation-in-the-loop ablation tested whether optimizing the schedule magnitude improves M0; Sec. VI-B summarizes the

negative result. M0 is retained in all reported system-level experiments.

E. Why the first mass moment is the essential state

Estimating CoM offset alone is ill-conditioned at release: physically $s = m_P r_{xy} \rightarrow 0$, whereas r_{xy} itself is undefined as $m_P \rightarrow 0$. In an earlier mass-only formulation, the optimizer could suppress a phantom eccentricity only by driving total mass below the bare-airframe value; 14 of 55 post-release samples then pinned at the lower bound. Augmenting s supplies the rotational channel directly and permits $c_{xy} = s/m$ without knowing the grasp-dependent horizontal offset. A frozen-window inertia coefficient prevents the optimizer from using instantaneous mass as a free inertia knob, while the weak r_z prior retains the dominant parallel-axis term. This structure is the minimum one that simultaneously represents payload existence, trim moment, and roll/pitch inertia without an external attachment geometry measurement.

The release detector studies three physically distinct sources: (i) collapse of $\rho = \|s\|/\|s\|_{\text{ref}}$ below 0.30, where the loaded reference is formed from a healthy, low-dynamic, in-envelope interval and frozen before aggressive flight, with an absolute $\|s\|$ bound used only as a sanity check; (ii) payload quantity, in which mass and algebraically derived inertia count as *one* source; and (iii) a directional thrust-residual edge with finite validity. Within this unified detector a single source never releases. The slow path requires quantity-empty and moment-collapse evidence for five MHE frames. The release-only residual is deliberately separate from the 4.4 N schedule-confirmation threshold and uses a signed two-window difference, $\Delta T = \bar{T}_{\text{post}} - \bar{T}_{\text{pre}}$, with 0.3 s half-windows. We evaluated two residual branches: fastA paired a normal 0.9 N residual with quantity-empty evidence, while fastB paired a longer-persistent residual with moment collapse. A recorded figure-eight transient reached -1.69 N, larger in magnitude than the -1.47 N step of the minimum supported payload, so amplitude alone cannot separate the cases. Online and offline evaluation call the same pure decision function. The randomized smoke of Sec. VI-C rejects fastA; the unified detector therefore fires only on moment-collapse evidence (slow or fastB) and treats an unresolved timeout as unknown, never as proof of an empty vehicle. Two further routes remain in the final interface and are reported with their frequencies in Sec. VI-C: a mass-domain path counted only in low-dynamic flight, and controller confirmation on the published no-payload confidence. Moment evidence must also originate from a fresh, successful MHE solve: the re-anchoring routine deliberately initializes s to zero after five consecutive solver failures, so that initialization is invalid evidence rather than a physical collapse.

V. SIMULATION SETUP

PX4 SITL with Gazebo [23] (Harmonic), an x500 quadrotor, and a magnetic DetachableJoint gripper that produces genuine inertia and CoM changes.

Two payload-change mechanisms, by design. The physical DetachableJoint gripper attaches a separate rigid body and therefore couples mass, inertia and CoM together; it carries every system-level claim (Tables IV, V, and II). Separately, a *wrench-emulated* step—a virtual force injection that changes the effective translational mass instantaneously, with no inertia or CoM change—is used only where a controlled ablation demands it: isolating the event trigger’s estimation-layer benefit requires a mass discontinuity that is clean and repeatable, free of the grasp-depth scatter and inertia mismatch that a physical attach necessarily introduces (Table I and Fig. 2 only). The supporting weight-schedule ablation runs on the physical gripper, as do all baseline and wind results. The two mechanisms are complementary, not interchangeable, and we keep their scopes disjoint: any conclusion that needs genuine rigid-body coupling is drawn only from the physical experiments (see the red-line note in §VI-A). The wrench mechanism’s inability to change rotational inertia is a documented simulator limitation, not a modeling choice; we state it and its consequences in §VII.

Task state machine: approach (position control) $\rightarrow$ NMPC hand-off and warmup hover $\rightarrow$ controlled descent $\rightarrow$ active grasp enable $\rightarrow$ lift $\rightarrow$ figure-eight $\rightarrow$ phase-triggered release. We state the NMPC scope explicitly: it governs the grasp hover and the post-event adaptation phase; the approach controller is not part of the claim. Headless batch infrastructure with a GCS-heartbeat stand-in; the physical-effectiveness timestamp $t = 0$ is defined by T_{phys} crossing the baseline, and in-band is defined as ± 0.08 kg.

Real-time budget. Both optimizers are acados [24] SQP solvers with merit backtracking and the HPIPM [25] QP solver. On one desktop core shared with Gazebo/PX4 SITL, the deployed 13-state NMPC ($N = 20$, 20 Hz) solves in 1.3 ms at the median and 2.2 ms at the 99th percentile ($n = 1009$), i.e. 2.6 % and 4.4 % of its 50 ms period. The deployed 16-state first-moment MHE ($N = 20$, 10 Hz) solves in 9.2 ms at the median, 24.6 ms at the 90th and 59.2 ms at the 99th percentile, with a 72.8 ms maximum ($n = 3740$, first 20 frames per flight discarded), against a 100 ms period—roughly $7\times$ the 1.4 ms median of the historical mass-only estimator. This meets the deadline in simulation, but the tail margin is narrow, so we make *no* onboard-headroom claim; that requires measurement on the target computer. Setpoints are published at 50 Hz by interpolating the latest prediction.

VI. RESULTS

A. Event-trigger benefit (estimation and control layers)

Table I (wrench-emulated pure mass step, $n = 6$). The event trigger cuts settling $\sim 4\times$ ($\delta m = -0.3$ kg) / $\sim 2.8\times$ (-0.8 kg); the **no-signal self-trigger recovers the benefit within noise** (~ 0.02 s of the signal version). Control-layer transient peak drops 6 % to 14 %.

Red line. This experiment reports settling and pos_err peak only; overshoot depth is a wrench inertia-mismatch artifact and is not reported here (overshoot conclusions come only from the physical-rigid-body slung experiments).

TABLE I

EVENT-TRIGGER VS. FIXED WEIGHTS (WRENCH PURE MASS STEP, $n = 6$).

δm	metric	fixed	signal	no-signal
-0.3	settling [s]	1.20	0.30	0.32
	pos_err peak [m]	0.200	0.187	0.173
-0.8	settling [s]	1.57	0.56	0.57
	pos_err peak [m]	0.545	0.483	0.482

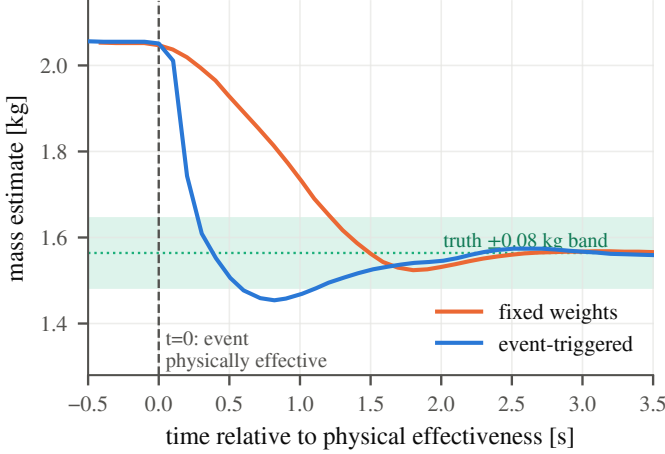


Fig. 2. Mass-estimate timeline (wrench step, $\delta m = -0.5$ kg). The event-triggered estimate reaches the truth band far earlier than fixed weights, at the cost of a transient overshoot below the band before settling—the speed/overshoot trade-off reported honestly.

B. Supporting schedule-tuning ablation

We additionally tested whether the handcrafted transient schedule required numerical optimization. A simulation-in-the-loop cross-entropy search [26] embedded M0 in a five-parameter family and evaluated 96 closed-loop episodes. The surviving candidate was then compared with M0 in a pre-registered paired 2×2 factorial ($n = 8$ per cell). Its median settling-time gain was -0.05 s for a 0.2 kg payload and 0.00 s for a 0.3 kg payload (both non-significant); the test had 97 % power against the larger previously observed effect. We therefore retain M0. This ablation supports a deliberately narrow conclusion: for the tested family, *when* to discard stale stages matters more than the precise weight magnitude. It is not a learning contribution of the deployed system.

C. Eventless grasp-release lifecycle

The controller issues the gripper-release command itself, but deliberately does not use that command to clear its payload model: a commanded release is not proof of physical detachment. We compare an event-driven operational reference (arm A), which resets payload geometry and inner-loop scaling with the command, against the proposed interface (arm B), which consumes only healthy, fresh $m/s/J$ frames. Arm B completes a release only after its no-payload confidence remains above 0.90 for 1.0 s. The two arms were paired under three payload and trajectory conditions. A pre-registered pair was voided and re-flown if either arm crashed before release or

TABLE II

EVENTLESS RELEASE COMPLETION AND CONFIRMATION DELAY (ARM B, VALID PAIRS ONLY). DELAY RUNS FROM THE RELEASE COMMAND TO THE FRAME AT WHICH THE NO-PAYLOAD CONFIDENCE HAS HELD ABOVE THRESHOLD FOR THE FULL 1.0 s HOLD, SO IT INCLUDES THAT HOLD BY CONSTRUCTION. THE TWO DELAY COLUMNS ARE THE TWO CONTROL PATHS DESCRIBED IN THE TEXT AND ARE *not* TO BE POOLED INTO A SINGLE MEDIAN.

condition	n	compl.	delay median [IQR]	
			in-maneuver	via brake-to-hover
W3: 0.15 kg, 4 m/s	6	6/6	2.78 s [2.76, 2.80]	16.53 s [16.46, 16.60]
W4: 0.30 kg, 4 m/s	7	7/7	7.35 s [7.20, 7.50]	16.40 s [15.76, 16.60]
W5: 0.15 kg, 2 m/s	8	8/8	3.76 s [3.16, 4.46]	16.45 s [16.40, 16.56]
pooled	21	21/21	3.76 s [2.86, 7.20]	16.46 s [16.36, 16.60]

Pooled completion 100 %, two-sided 95 % Clopper-Pearson CI [83.9 %, 100 %]; overall delay max 16.60 s. The share of flights confirming only after the brake-to-hover maneuver is 67 % (W3), 71 % (W4), 50 % (W5) and 62 % pooled; the complement confirms in maneuver.

lacked payload evidence; these exclusions are reported below and are not absorbed into the completion result.

Among valid pairs, arm B completed autonomous release confirmation in 21/21 flights with no observed post-release instability (Table II). The two-sided 95 % Clopper-Pearson interval is [83.9 %, 100 %], so this result demonstrates closed-loop feasibility rather than a zero failure rate.

The confirmation delay is *bimodal*, and the two modes are two distinct control paths rather than a spread within one path. In 38 % of flights the no-payload confidence crosses threshold while the vehicle is still flying the figure-eight and confirmation completes in 3.76 s (median). In the remaining 62 % no moment vote is available in maneuver— $\hat{s}$ has drifted under sustained excitation or the frozen reference never formed—so the system reaches its 12 s timeout, declares the payload state UNRESOLVED, and executes a 3 s smooth brake to hover; deceleration alone restores observability and confirmation then completes at 16.46 s (median). Reporting one median across both paths would be misleading, so Table II separates them.

In 17 of the 21 flights the payload-presence machine made exactly two transitions—one load declaration at grasp and one empty declaration at release—with no spurious carrying-phase transition. In the four remaining flights, all of which confirmed in maneuver, the estimator’s presence machine never declared EMPTY within the logged window (over 40 s past confirmation); confirmation rested on the published no-payload confidence alone, as detailed below. Three of the 21 load declarations were raised by the estimator’s own thrust-residual self-trigger rather than by sustained mass evidence; no external grasp signal was consumed. The published first moment decayed continuously rather than stepping to zero, confirming that recovery used the intended estimator interface rather than the controller’s release command.

a) *Safety boundary*: The 21/21 batch validates the interface once its release evidence is accepted; it does not validate a universally robust detector. We therefore tested three physically distinct evidence sources: payload quantity, directional thrust residual, and collapse of the normalized

TABLE III
TARGETED RELEASE-SAFETY ABLATION.

decision rule	flights	premature	confirmed / unresolved
quantity + residual fast path	9	3	—
fresh moment-gated paths	12	0	11 / 1

first moment. A fast branch combining quantity and residual without moment evidence declared release prematurely in 3/9 randomized flights during a 4 m/s figure-eight. Loaded-flight mass estimates could remain below the nominal empty threshold for up to 48 s, while a dynamic thrust excursion reached -1.69 N, exceeding the -1.47 N weight change of the smallest payload. Neither source is therefore reliable on its own in this regime.

We rejected that branch: the estimator’s unified release detector now fires only on fresh, successful moment-collapse evidence (the slow and fastB paths); the legacy mass-domain route that remains is quantified below. Ambiguous cases enter UNRESOLVED; a timeout does not clear the model or reset adaptive state. In a subsequent 12-flight smoke test, no flight declared release before the command: all 11 confirmed releases passed through moment-gated paths (fastB 9, slow 2) with a median detector delay of 1.9 s, and one remained safely unresolved (Table III). That smoke preceded the frozen moment reference.

b) Release paths in the final batch: The moment-gated detector is not the only route to confirmation in the final code, and the 21-flight batch of Table II exercised three. The controller confirms release when the published no-payload confidence—the product of mass, first-moment and inertia “empty” scores, where the moment score uses an absolute bound on $\|s\|$ rather than the frozen-reference ratio—holds above 0.90 for 1.0 s on healthy, fresh frames. (i) In 12/21 flights (W3 6/6, W5 6/8) the moment-gated detector declared release first and the confidence followed. (ii) In 5/21 flights, the five W4 flights that reached UNRESOLVED, the frozen moment reference was never established because its low-dynamic, in-envelope sampling window never filled, so the ratio test never ran; the confidence crossed threshold during or shortly after the brake to hover, and the estimator’s mass-domain path ($m_P < 0.030$ kg for 20 frames, counted only in low-dynamic flight) declared release 0.9–3.1 s later. (iii) In 4/21 flights (two in W4, two in W5), all confirmed in maneuver, the reference was likewise never established and the estimator never declared release; confirmation rested on the confidence alone. No flight in any path declared release before the command. The supported safety conclusion is therefore bounded: the smoke validates the moment-gated detector, while the final batch shows that 9/21 confirmations did not pass through it. Broader detector robustness remains outside the present claim.

The paired performance comparison is conditional on successful attachment and survival to the release command. Of 77 flights, 20 were voided and re-flown: 17 crashed before release and three lacked MHE payload evidence. The available

TABLE IV
THREE-WAY METHOD $\times$ SCENARIO MATRIX ($n = 5$): POS_ERR PEAK [M] / RECOVERY [S].

cell (mass/ecc)	truth	online	l1
0.2 / 0.05	0.806 / 4.28	0.768 / 3.70	0.821 / 4.26
0.2 / 0.10	0.794 / 3.62	0.810 / 3.92	0.812 / 4.24
0.3 / 0.05	0.834 / 3.56	0.837 / 3.72	0.883 / 4.90
0.3 / 0.10	1.011 / 3.64	0.876 / 3.88	0.891 / 4.54

logs cannot distinguish physical capture failure from estimator non-recognition in those three cases. Among valid pairs, the tail-window tracking RMS is no longer comparable between the arms under the current code: in the 62 % of arm B flights that reach UNRESOLVED, the final 20 s are flown at hover while arm A continues the figure-eight, which accounts for the entire apparent difference. We therefore draw no steady-tracking comparison from this batch. The measurable cost of refusing the event shortcut is the confirmation delay: a median 3.76 s when the moment vote clears in maneuver, and 16.46 s when it does not and the vehicle must first brake to hover.

D. Baselines and ablations

L1-NMPC three-way matrix (Table IV). {truth, online, l1} $\times$ {0.2, 0.3} kg $\times$ {0.05, 0.10} m, $n = 5$, 60/60 runs stable, zero divergence. (i) Transient peaks are statistically indistinguishable across the three methods (0.77 m to 1.01 m, within-cell differences 0.01 m to 0.09 m)—L1 is a competent strong baseline and our method is not worse on the transient. (ii) **L1 recovers 15 % to 35 % slower**; the mechanism is that its compensation low-pass cutoff ω_c is bounded by closed-loop stability (0.5 stable, 1.0 unstable on this platform), a bandwidth wall that explicit estimation does not face. (iii) L1 produces no interpretable m/c_{xy} (cannot update thrust-margin constraints), has no estimation-layer settling notion, and still requires the same class prior for gain scaling. Implementation (footnote): L1 is a translational 3-DoF gradient-law augmentation using T_{phys} as the observed thrust; using the commanded thrust self-excites, and a ground low-thrust gate is required.

Wind false-trigger ablation (Table V). Self-trigger false-positive rate vs. sustained wind magnitude (link-wrench injection, no mass event, $n = 3$). (i) Sharp step at the 1.5 N threshold: sub-threshold vertical wind never false-triggers, supra-threshold always does—a sustained vertical force is **inherently indistinguishable from a mass change on the thrust-residual channel** (an honest limitation). (ii) **Immune to horizontal wind**: even at $2\times$ threshold (3 N) horizontal wind never false-triggers, because the detector lives on the collective-thrust channel. (iii) Mitigations: the persistence gate rejects gust pulses; real wind is predominantly horizontal/gusty rather than sustained-DC vertical; the threshold is tunable to the noise floor.

VII. LIMITATIONS AND FUTURE WORK

All system-level evidence is presently SITL-only. Hardware migration replaces simulated motor speeds with ESC telemetry

TABLE V
WIND FALSE-POSITIVE RATE VS. MAGNITUDE ($n = 3$, NO MASS EVENT).

force [N]	0.5	1.0	1.5	2.0	2.5	3.0
vertical FPR	0/3	0/3	3/3	3/3	3/3	3/3
horizontal FPR	—	0/3	—	0/3	—	0/3

and must re-identify thrust/torque coefficients, noise floors, timing, and gripper compliance before free flight. The vertical arm r_z is unobservable from the thrust-moment channel and remains a weak rack prior; the s_x /pitch channel is noisier than s_y /roll. The deployed diagonal inertia approximation also drops the small $O(r_{xy}^2)$ and yaw-inertia terms.

Release-detector boundary. First-moment augmentation is now part of the default path. The 21/21 paired result shows that the first-moment output can drive complete NMPC recovery without a drop event, but it does not establish distribution-wide detector robustness. The targeted safety ablation identifies the operative boundary: payload quantity and thrust residual can become simultaneously false-empty during dynamic flight, causing premature model clearing in 3/9 flights when combined without moment evidence. The accepted unified detector therefore fires only on fresh, healthy moment-collapse evidence and treats timeouts as UNRESOLVED, never as proof of an empty vehicle. After this correction, 12 flights produced no premature declarations, 11 confirmations, and one safe unresolved outcome. In the final 21-flight batch, however, only 12 confirmations passed through that detector: five followed the mass-domain path after a brake to hover and four rested on the no-payload confidence alone, because the frozen moment reference was never established in those flights. None was premature.

The remaining detector limitation is its moment reference. A running maximum absorbs dynamic excursions above the feasible envelope and makes an ordinary loaded moment appear collapsed, so the deployed reference is formed only from a healthy, low-dynamic, in-envelope interval and frozen before aggressive flight. That interval must follow the LOADED declaration; in 9/21 final-batch flights it never filled, so the ratio test never ran. Until reference formation is reliable, our claim is an eventless, fail-safe control interface rather than a fully robust autonomous release detector.

The current software also has a configuration-coupling defect: the historically named `c_xy_est_enable` parameter is the effective master switch for all release evidence. Although every reported release batch explicitly enabled it, the bare default is `false` and therefore falls through to UNRESOLVED. Before deployment, release decision execution must receive its own explicit enable parameter (or be unconditional in the eventless profile), with startup validation that rejects inconsistent configurations.

Excitation-induced drift of the first moment. The dominant limitation of the release detector is not threshold placement but an observability boundary. Under sustained figure-eight excitation the estimated first moment $\hat{s}$ drifts upward: in

affected flights the loaded $\|\hat{s}\|$ grows from 0.012 kg m to 0.031 kg m ($2.5\times$), and the frames that exceed the physical envelope recur with a 12.0 s period, matching the figure-eight half period $\pi/w = 11.1$ s, i.e. the spacing of the lateral-acceleration extrema. While drifted, the moment vote is correctly *withheld* rather than mis-cast, so the failure mode is a delayed confirmation, not a false release. The drift is reversible: deceleration alone restores $\hat{s}$, which collapses below the release threshold within 3.3 s to 3.4 s of leaving the maneuver. With the flights whose frozen reference never formed, 62 % of flights ($n = 21$) confirm only after the 12 s timeout and the 3 s brake-to-hover, giving a confirmation delay of 16.5 s instead of a few seconds.

We can detect the condition but cannot yet predict it. The count of envelope-exceeding frames (`ovr`), already computed online, separates the two populations at 4 m/s: among the 13 4 m/s flights, those with `ovr`=0 confirmed in maneuver in 3/3 cases versus 1/10 for `ovr`>0 (Fisher $p = 0.014$; pooled with 2 m/s, 7/11 vs. 1/10, $p = 0.024$), but all eight 2 m/s flights had `ovr`=0 and half of them still required the brake to hover, so `ovr` is not a sufficient indicator at low speed. What remains unexplained is *which* flights drift. We pre-registered the hypothesis that drift is driven by maneuver intensity and tested it against the 2 m/s condition, where the affected share should have dropped sharply; it did not (50 % at 2 m/s versus 67 % and 71 % at 4 m/s). We therefore report the intensity explanation as *not supported* and leave the trigger open. At 4 m/s, a gate on `ovr` would let the vehicle leave the maneuver as soon as the moment channel is known to be unusable, cutting the delay from 16.5 s to approximately 3.5 s; this is implemented in neither the estimator nor the controller and is not claimed here.

Selection and pre-release failures. The main performance tables condition on successful attachment and survival to the release command. Twenty of 77 flights were voided; arm A had 11/39 pre-release crashes and arm B had 6/38, while arm B alone had 3/38 failures of the pre-registered MHE payload-evidence check. The historical logging recorded an attach request but not the simulator joint-state acknowledgement, so those three cases cannot be separated into physical capture failures versus estimator non-recognition. Arm A crashes concentrate in W4 (six of eleven), which exhausted the pre-registered attempt cap at seven valid pairs there rather than the intended eight; W3 stopped at six pairs for scheduling reasons rather than by the cap. Consequently the paper makes no claim that the interface improves grasp, lift, or early figure-eight stability, and W3/W4 carry somewhat less power than W5. Attachment observability and the pre-release attitude instability require separate experiments rather than reinterpretation of the conditional drop result.

A sustained vertical up/downdraft remains intrinsically confounded with a payload change on a collective-thrust-only residual; the existing wind ablation characterizes this limitation but does not remove it.

Simulator fidelity. (i) gz-sim Harmonic cannot change a link's inertia at runtime in a way the physics adopts (issue

#2733), so the wrench mechanism emulates only a translational mass step; every inertia, CoM, or eccentricity claim comes from the DetachableJoint platform (§V). (ii) The payload is rigidly attached; gripper compliance and pendulum modes are not modeled. Neither affects which signals the estimator consumes.

On learning. The schedule search (Sec. VI-B) was bound by SITL sample economics: ~ 85 s per episode and no injectable random seed, so per-candidate noise matched the effect sought. Per-frame supervised targets or a lower-fidelity search environment remain open; the result does not argue against learned estimator tuning in general.

VIII. CONCLUSION

We presented an eventless payload-adaptive NMPC architecture that uses motor speeds and odometry to update mass, first mass moment, and inertia without consuming attach/drop notifications or treating a release command as proof of physical detachment. Self-triggered MHE window deweighting accelerates abrupt-change recovery, while the atomic, health-gated $m/s/J$ interface and continuous body-rate scheduling connect the estimate safely to the controller. Conditional on successful attachment and pre-release survival, the interface completed release in 21/21 paired SITL flights with no observed post-release instability. It also matched the transient tracking of an L1-NMPC baseline while recovering 15% to 35% faster and retaining interpretable physical parameters.

The release-safety ablation defines the claim boundary. Payload quantity and thrust residual became simultaneously false-empty during aggressive flight, so the accepted unified detector fires only on fresh moment-collapse evidence and preserves ambiguous cases as UNRESOLVED. This policy produced zero premature declarations in the corrected 12-flight smoke, with 11 confirmed releases and one safe unresolved outcome; in the final batch 12/21 confirmations passed through it, and the remaining nine, via a low-dynamic mass path or the no-payload confidence, were also never premature. The present evidence therefore demonstrates that the payload-event signal can be removed from the adaptive control interface; it does not establish a fully robust release detector. Reliable formation of the frozen moment reference, broader SITL validation, and subsequent HITL and flight testing are the next steps.

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